5 - Linear Regression I

From a Population Relationship to an OLS Line

Filipa Reis (filipareis@ucp.pt)

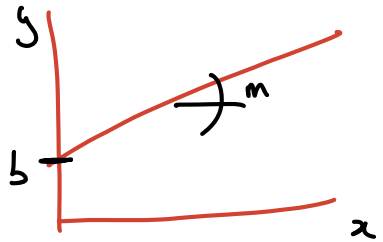
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Agenda

Today we move from estimating **one population quantity** to estimating a **relationship between variables**.

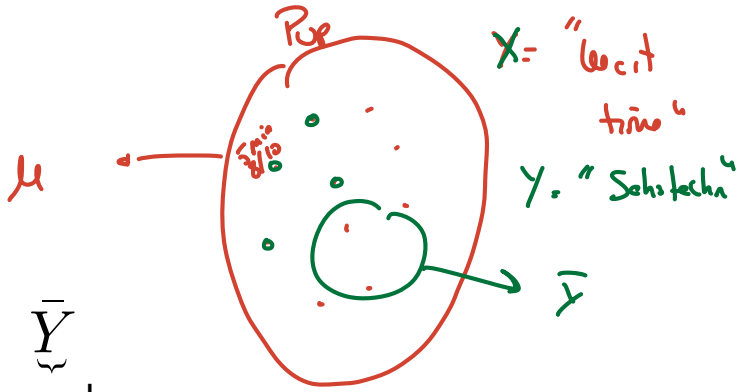
1. Population relationships and regression parameters
2. What the researcher actually observes: a sample
3. How Ordinary Least Squares (OLS) chooses a line
4. Goodness of fit and what regression does **not** tell us
5. Different samples produce different regression lines

Recommended reading: Wooldridge, Chapter 2.1–2.3



$$y = mx + b$$

Same Inferential Logic, New Parameters



So far, a typical problem looked like:

$$\underbrace{\mu}_{\text{unknown population parameter}} \longleftarrow \underbrace{\bar{Y}}_{\text{sample estimator}}$$

Regression asks about a different population feature:

$$\underbrace{\beta_0,\ \beta_1}_{\text{unknown population parameters}} \longleftarrow \underbrace{\hat{\beta}_0,\ \hat{\beta}_1}_{\text{sample estimators}}$$

The logic is the same: we use sample data to learn about an unknown population relationship.

1. From a Business Question to a Relationship

What Helps Explain Laptop Prices?

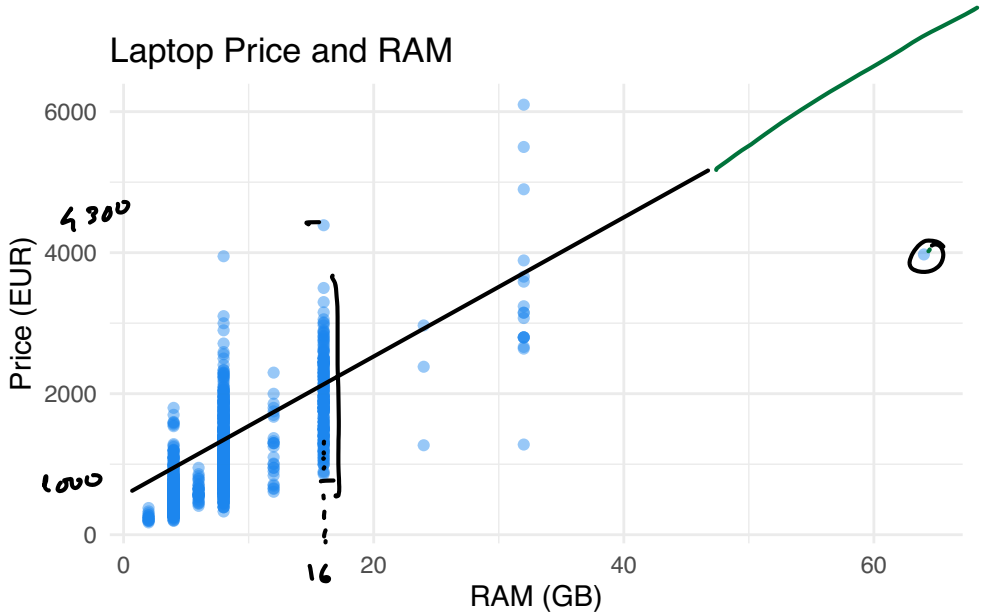
Y → outcome / dependent

X → explanatory / independent

Question: How is laptop price related to RAM?

Before fitting a model:

- What is the **outcome** we want to explain?
- What is the **explanatory variable**?
- What relationship do you expect?
- What other factors might also affect price?



→ Brand, storage, CPU, screen size, touch

The Population Relationship We Want to Learn

A simple linear regression model represents the population relationship as:

$$y = \beta_0 + \beta_1 x + u$$

For laptop prices:

$$Price = \beta_0 + \beta_1 RAM + u$$

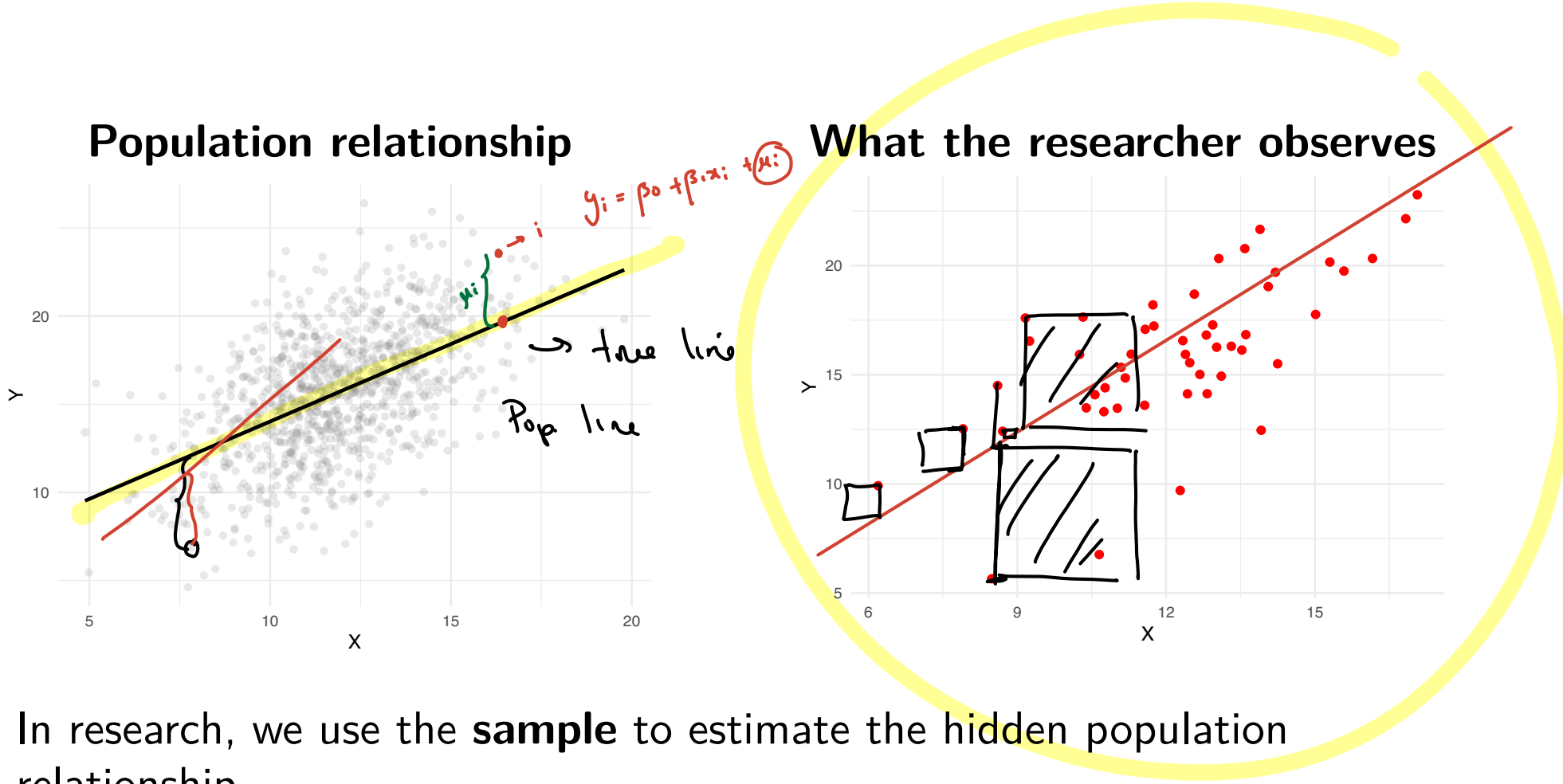
- β_0: population intercept
- β_1: population slope
- u: all other influences on Y not explicitly represented by X

For laptop prices, u may contain **brand, processor, screen, storage, build quality**, and other unobserved or unmodeled factors.

β_0 and β_1 are **population parameters**. We do not observe them directly.

2. Population vs. Sample

What We Want to Know vs. What We See



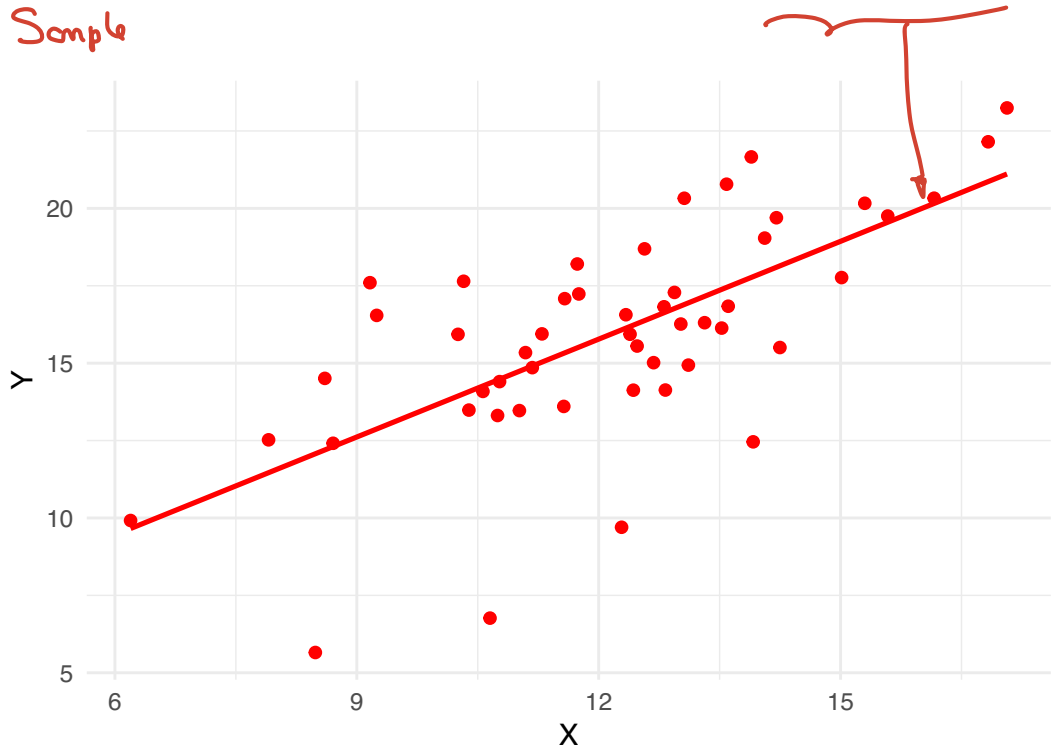
In research, we use the **sample** to estimate the hidden population relationship.

From the Sample to an Estimated Line

We use the sample data to estimate the unknown population relationship:

Population model $\longrightarrow$ $y = \beta_0 + \beta_1 x + u$

Estimated model $\longrightarrow$ $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$



3. Ordinary Least Squares

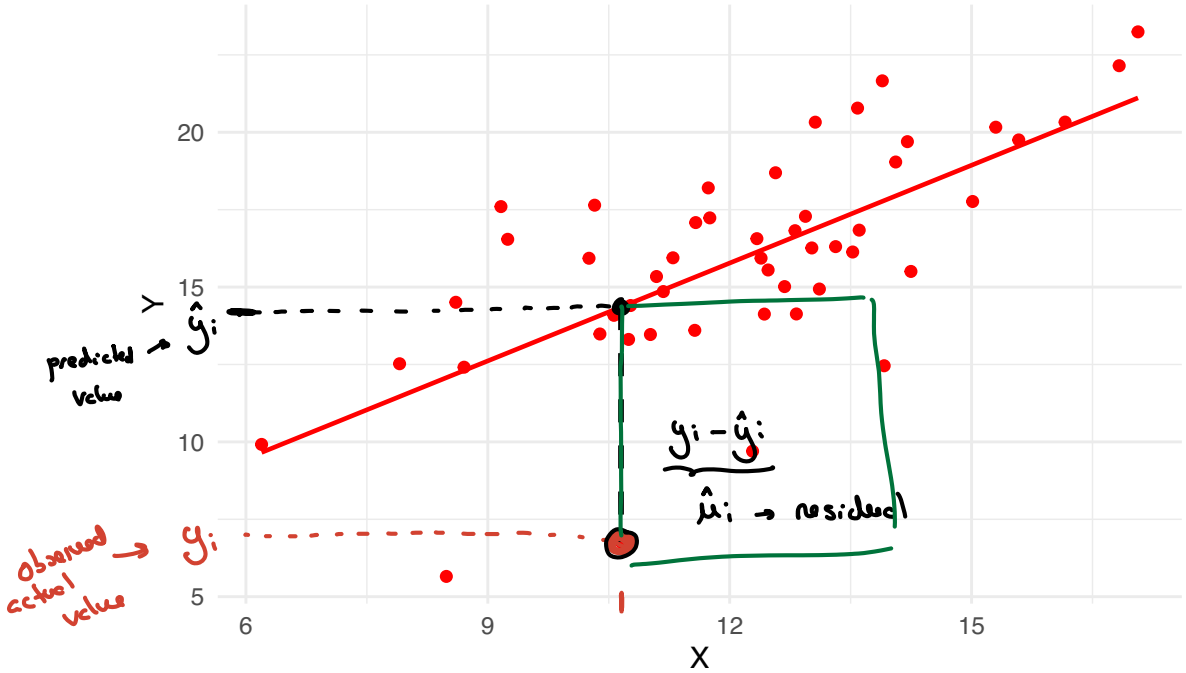
Actual Values, Fitted Values and Residuals

For observation (i): $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

is the **fitted** value. But we actually **observe** y_i

The **residual** is the vertical gap between the observed and fitted values:

$$\hat{u}_i = y_i - \hat{y}_i.$$



How Does OLS Choose the Line?

There are infinitely many possible lines through the cloud of points.

Ordinary Least Squares (OLS) chooses $\hat{\beta}_0, \hat{\beta}_1$ to minimize:

$$\min \sum_{i=1}^{n} \hat{u}_i^2 = \sum_{i=1}^{n} \left[y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right]^2$$

So OLS minimizes the **sum of squared residuals**.

Why square the residuals?

- positive and negative residuals cannot cancel;
- large misses receive more weight;
- the criterion gives a unique, tractable solution under ordinary conditions.

The Sample Regression on Laptop Data

Now fit the model using the observed laptop data:

$$\widehat{Price}_i = \hat{\beta}_0 + \hat{\beta}_1 RAM_i$$

```
# Estimate the simple linear regression
m1 <- lm(Price_euros ~ Ram, data = laptops)
# Coefficient estimates
coef(m1)
```

```
(Intercept)         Ram
   276.0273    101.7609
```

For these data:

$$\widehat{Price}_i = 276 + 101.8 \cdot RAM_i$$

Interpreting the Estimated Coefficients

Slope: $\hat{\beta}_1 = 101.8$

- According to our fitted model, laptops with **1 additional GB of RAM** are associated with approximately **€102 higher price**, on **average**.

Intercept: $\hat{\beta}_0 = 276$

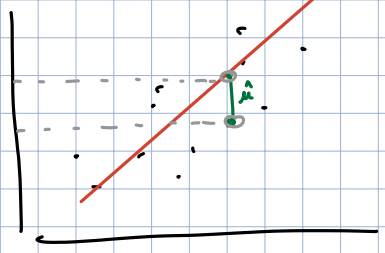
- The intercept is the **estimated average price when RAM = 0 GB**.
- Here, the intercept has little practical meaning because a laptop with **0 GB of RAM** is not a realistic case.
- Note that an intercept may have little substantive meaning when $X = 0$ is unrealistic or outside the relevant data range.

Linear Regression

$$y = \beta_0 + \beta_1 x_i + \mu$$

$$Price = \beta_0 + \beta_1 RATI + \mu \rightarrow \text{Pop. model}$$

$$\hat{Price} = \hat{\beta}_0 + \hat{\beta}_1 RATI \rightarrow \text{Sample / Estimated model}$$



OLS

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum \hat{\mu}_i^2$$

$$\hat{\mu}_i = y_i - \hat{y}_i$$

What SLR Does and Does Not Hold Constant

A simple regression compares laptops with different RAM values.

It does **not** automatically compare laptops that are identical in:

- brand;
- CPU;
- storage;
- screen;
- build quality;
- other characteristics.

Therefore: $\hat{\beta}_1$ summarizes the **sample association** between RAM and price.

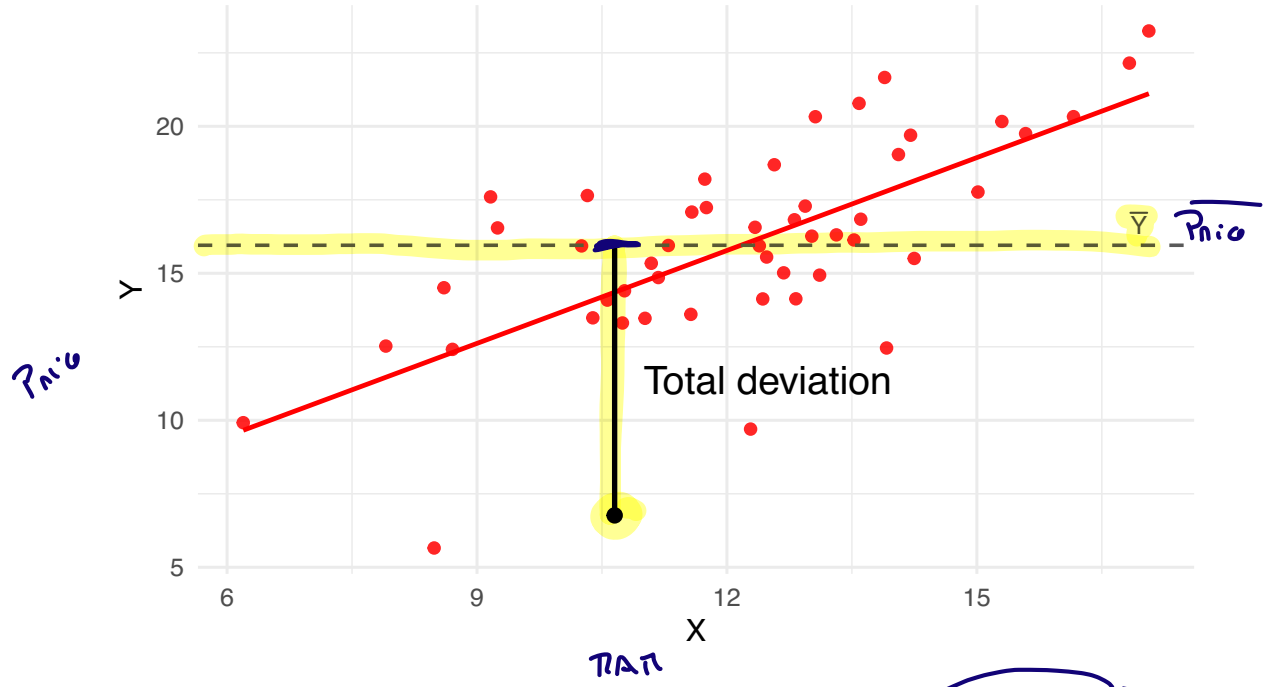
A causal interpretation requires additional assumptions that we will study later.

4. How Well Does the Line Fit This Sample?

From Residuals to Model Fit

Before using RAM, the simplest benchmark is to predict the **same price for every laptop**:

$$\hat{y}_i = \bar{y}$$



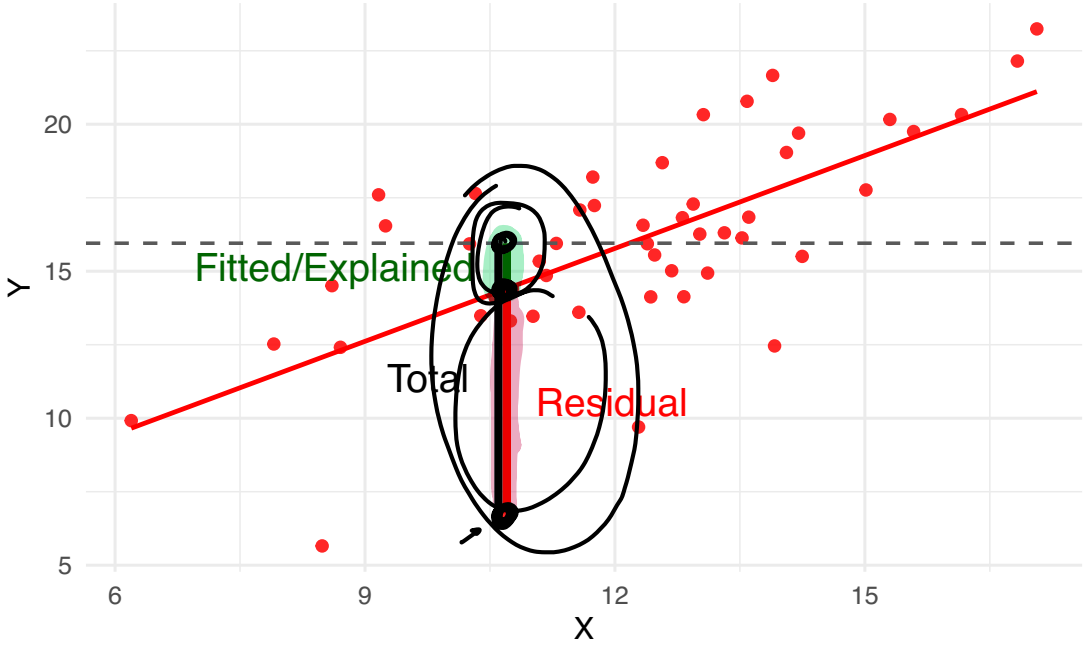
For one laptop, its deviation from the mean is: $y_i - \bar{y}$

Across the sample, these deviations describe the **total variation in prices**.

Decomposing One Observation's Deviation

For the same observation, the deviation from the mean can be split into:

- a part captured by the fitted line;
- a part left over as a residual.



$$\underbrace{y_i - \bar{y}}_{\text{total deviation}} = \underbrace{\hat{y}_i - \bar{y}}_{\text{fitted part}} + \underbrace{y_i - \hat{y}_i}_{\text{residual part}}$$

Two Ways to Describe Model Fit

The variation left over after fitting the regression is:

$$\sum_i \hat{u}_i^2$$

Both R^2 and the SER start from these same residuals, but answer different questions.

R^2: Relative Fit

$$R^2 = 1 - \frac{\sum_i \hat{u}_i^2}{\sum_i (y_i - \bar{y})^2}$$

How much of the variation in y does the model account for?

- A **proportion** (between 0 and 1)

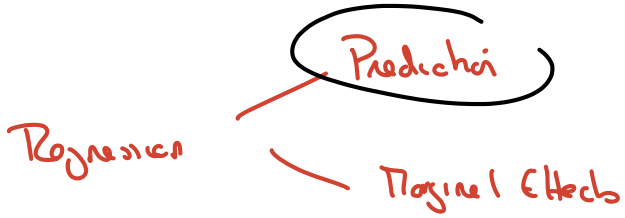
SER: Residual Scale

$$\hat{\sigma} = \sqrt{\frac{\sum_i \hat{u}_i^2}{n-2}}$$

How large is the remaining variation around the fitted line?

- Same units as Y (here in euros)

Reading R^2 and SER Together



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For the laptop regression:

$$R^2 = 0.55 \qquad SER = €471$$

- R^2: RAM accounts for about **55% of the variation** in laptop prices.
- **SER:** the typical vertical distance between observed laptop prices and the fitted regression line is about **€471**.

$$R^2 \longrightarrow \text{How much variation is accounted for?}$$
$$SER \longrightarrow \text{How far do observations typically fall from the line?}$$

Important: Better fit does not by itself imply a more credible or causal model.

A Good Fit Is Not the Same as a Good Explanation

A high R^2 does **not** tell us that:

- the model is correctly specified;
- RAM causes price;
- omitted factors are unimportant;
- the comparison is fair.

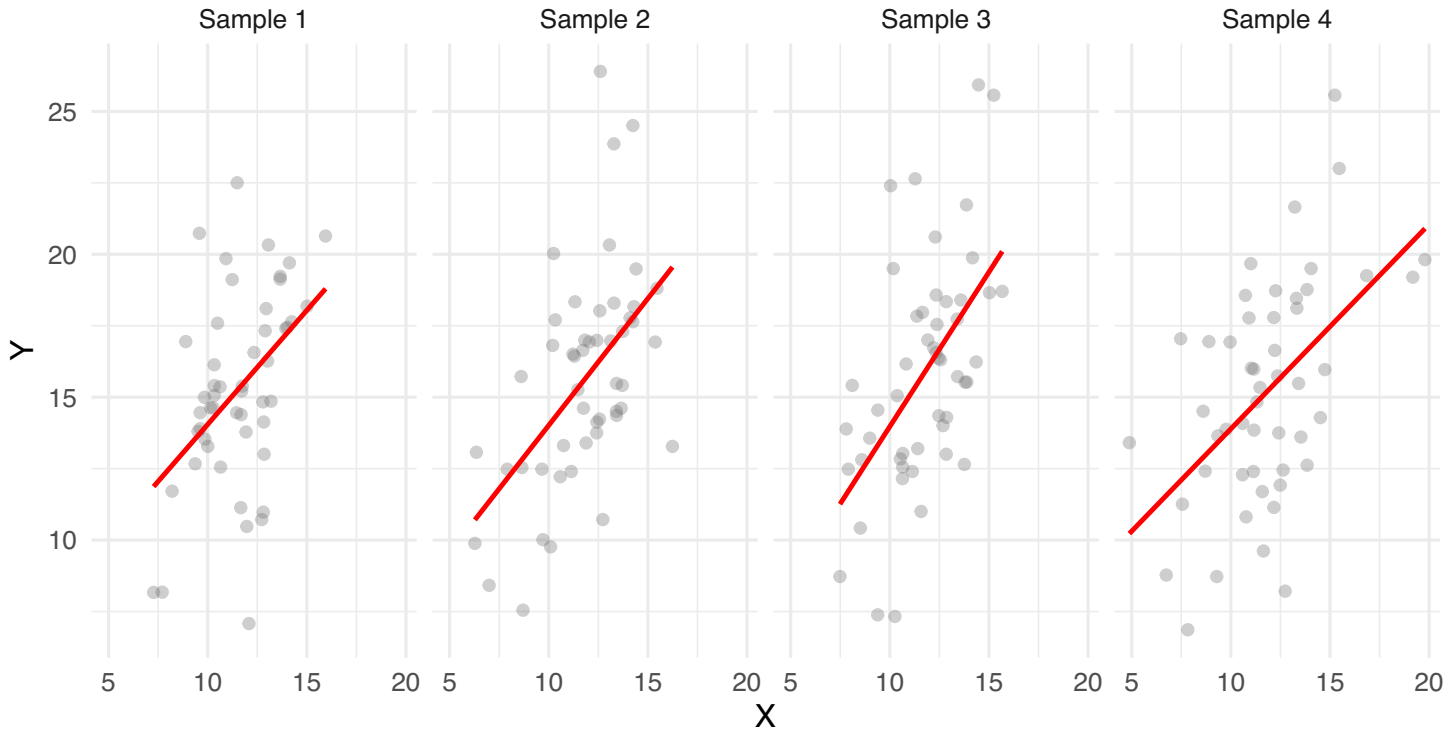
A low R^2 does **not** automatically make a relationship unimportant.

Regression gives us a **disciplined summary of a relationship**. What we are allowed to conclude depends on the question, the data, and later, the assumptions.

5. Different Samples, Different Regression Lines

The Estimated Line Depends on the Sample

If we took another random sample from the same population, OLS would generally give us a different line.



The **population parameters** β_0, β_1 are fixed.

The **estimates** $\hat{\beta}_0, \hat{\beta}_1$ vary from sample to sample.

6. Closing the First Regression Lecture

Key Takeways

Population relationship $\rightarrow$ random sample $\rightarrow$ OLS $\rightarrow \hat{\beta}_0, \hat{\beta}_1$

$\hat{\beta}_0, \hat{\beta}_1 \rightarrow$ fitted values + residuals $\rightarrow$ model fit

Remember:

- β_0, β_1 are **population parameters**; $\hat{\beta}_0, \hat{\beta}_1$ are **sample estimates**.
- OLS chooses the line that minimizes the **sum of squared residuals**.
- $\hat{\beta}_1$ describes a **linear association**; it is not automatically a causal effect.
- R^2 and SER describe **sample fit**, not causal credibility.

Appendix / Reference

In simple regression with an intercept:

$$\hat{\beta}_1 = \frac{\text{Cov}(x, y)}{Var(x)} = r_{xy} \frac{s_y}{s_x}$$

So the slope depends on:

- the **direction and strength** of the linear association;
- the **units / scale** of X and Y.

Correlation is unit-free. The regression slope is not.

OLS Closed-Form Estimates

For simple regression with an intercept:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The fitted OLS line therefore passes through $(\bar{x}, \bar{y})$.

Variation Decomposition and R^2

With an intercept:

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{Total variation}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{Explained variation}} + \underbrace{\sum_i \hat{u}_i^2}_{\text{Residual variation}}$$

and:

$$R^2 = \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{\sum_i \hat{u}_i^2}{\sum_i (y_i - \bar{y})^2}$$

Readings

- Wooldridge, *Introductory Econometrics: A Modern Approach*, 7th ed., Chapter 2
  - 2.1 Definition of the Simple Regression Model
  - 2.2 Deriving the Ordinary Least Squares Estimates
  - 2.3 Properties of OLS on Any Sample of Data